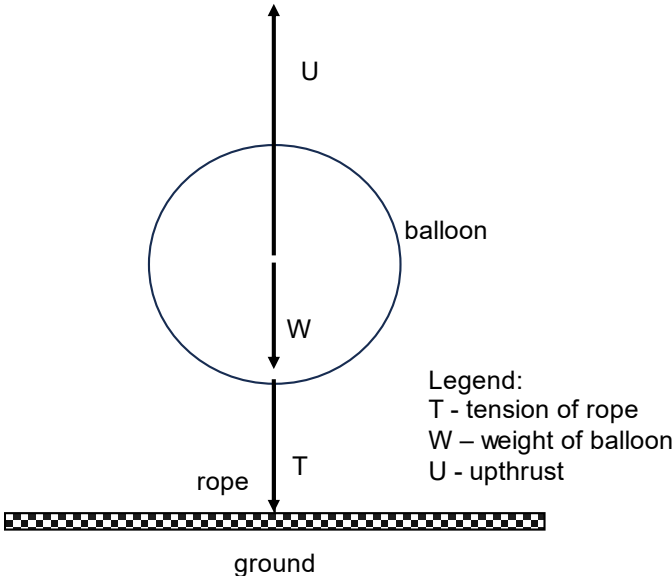


3(a)(i)	It is the resultant vertical upward force exerted by the fluid on an object as the pressure at the bottom surface is greater than that at the top. Upthrust is equal in magnitude and opposite in direction to the weight of the fluid displaced by a submerged or floating object.
3(b)(i)	 B1 – correct direction with labeling B1 – correct length of U, W and T and point of action
(ii)	$U = \rho Vg$ $= (1.3)\left(\frac{4}{3}\pi\left(\frac{1.5}{2}\right)^3\right)(9.81) = 22.54N$
	$U = T + mg$ $T = U - mg$ $= 22.54 - (0.3)(9.81) = 19.6N$
(b)(ii)	Net force upwards as upthrust is more than weight
	As altitude increases, air pressure/density decreases, causing upthrust to decrease
	Upthrust will decrease till its magnitude is equal to weight.